

Newton's Method

A Second-Order Optimization Algorithm

Derivation · Convergence Analysis · Riemannian Perspective

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CS 435 — Algorithms for Convex Optimization
Yale University, 2018

April 12, 2018

Lecture Outline

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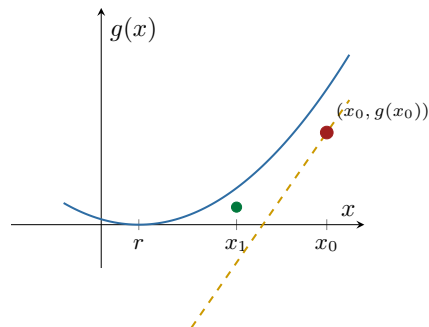
Newton-Raphson Method: Root Finding

Problem Statement

Given $g : \mathbb{R} \rightarrow \mathbb{R}$, find $r \in \mathbb{R}$ such that $g(r) = 0$.

Geometric Idea

- Start at x_0 near a root
- Draw the **tangent line** at $(x_0, g(x_0))$
- Let $x_1 =$ intersection with the x -axis
- Repeat: each step brings us *closer* to the root



Tangent at x_0 intersects x -axis at x_1 , closer to root r .

Update Rule

$$x_{k+1} := x_k - \frac{g(x_k)}{g'(x_k)}, \quad k \geq 0$$

Requires: $g \in C^2$ (twice continuously differentiable)

Example: Computing $1/a$ Without Division

Setup

Minimise $f(x) = ax - \log x$ over $x > 0$.

$$g(x) = f'(x) = a - \frac{1}{x} \Rightarrow r = \frac{1}{a}$$

Newton Iteration

$$x_{k+1} = 2x_k - ax_k^2$$

No division required! (only $+$ and $\times$)

Historical note: IBM used this trick in hardware.

Convergence Analysis

Let $e_k := 1 - ax_k$. Then:

$$e_{k+1} = e_k^2$$

- $|e_0| < 1$: $e_k \rightarrow 0$ ✓
- $|e_0| = 1$: $e_k = 1$, stuck
- $|e_0| > 1$: $e_k \rightarrow \infty$ ✗

In terms of x_0 : converges iff $0 < x_0 < \frac{2}{a}$

Key Lesson

The **initial point** is critical for Newton's method!

Theorem 1: Quadratic Convergence

Theorem (Quadratic Convergence for Root Finding)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be C^2 , r a root, x_0 a starting point, $x_1 = x_0 - g(x_0)/g'(x_0)$. Then:

$$|r - x_1| \leq M |r - x_0|^2, \quad M = \sup_{\xi \in [r, x_0]} \left| \frac{g''(\xi)}{2g'(x_0)} \right|$$

Proof Sketch

- ① Taylor-expand $g(r) = 0$ at x_0 (MVT):

$$0 = g(x_0) + (r - x_0)g'(x_0) + \frac{1}{2}(r - x_0)^2 g''(\xi)$$

- ② Use $g(x_0) = g'(x_0)(x_0 - x_1)$
③ Algebra gives $g'(x_0)(r - x_1) = \frac{1}{2}(r - x_0)^2 g''(\xi)$
④ Divide by $|g'(x_0)| \Rightarrow$ result

Convergence Rate

If $M \leq 1$ and $|x_0 - r| < \frac{1}{2}$:

$$|x_k - r| \leq |x_0 - r|^{2^k} \leq 2^{-2^k}$$

To reach error ε :

$$k \approx \log \log \frac{1}{\varepsilon} \text{ steps}$$

Extension to Multivariate Functions

Setting

Find $x \in \mathbb{R}^n$ satisfying the system

$g_1(x) = \dots = g_n(x) = 0$, i.e. $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$,
 $g(x) = \mathbf{0}$.

Jacobian Replaces Derivative

The Jacobian $J_g(x_0) \in \mathbb{R}^{n \times n}$:

$$g(x) = g(x_0) + J_g(x_0)(x - x_0) + o(\|x - x_0\|^2)$$

$$[J_g(x_0)]_{ij} = \frac{\partial g_i}{\partial x_j}(x_0)$$

Multivariate Update Rule

$$x_{k+1} := x_k - J_g(x_k)^{-1} g(x_k), \quad k \geq 0$$

Division $\rightarrow$ **matrix inversion**

Convergence

Local quadratic convergence still holds. The constant M now involves:

- Upper bound on Lipschitz const. of J_g
- Lower bound: $1 / \|J_g(x)^{-1}\|$

From Root-Finding to Optimization

Key Bridge

For smooth convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$x^* = \operatorname{argmin}_{x \in \mathbb{R}^n} f(x) \iff \nabla f(x^*) = \mathbf{0}$$

Apply Newton's method to $g := \nabla f$!

Newton Step & Update

Since $J_{\nabla f}(x) = \nabla^2 f(x)$:

$$x_{k+1} := x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

Define the **Newton step** at x :

$$n(x) := -(\nabla^2 f(x))^{-1} \nabla f(x)$$

Second-Order Interpretation

Minimise the local **quadratic approximation**:

$$\tilde{f}(x) = f(x_0) + \langle x - x_0, \nabla f(x_0) \rangle + \frac{1}{2} (x - x_0)^\top \nabla^2 f(x_0) (x - x_0)$$

Setting $\nabla \tilde{f}(x) = 0$ yields *exactly* the Newton step!

$\Rightarrow$ Newton's method = minimise a fresh quadratic at each step.

Computational Cost / Iteration

Solve $\nabla^2 f(x_k) d = -\nabla f(x_k)$:

- General: $O(n^3)$ (Gaussian elimination)
- Structured (SDD, Laplacian): $\tilde{O}(m)$

Example: Logarithmic Barrier & Analytic Center

Polytope & Logarithmic Barrier

$$P = \{x \in \mathbb{R}^n : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$$

$$F(x) = - \sum_{i=1}^m \log(b_i - a_i^\top x)$$

Each term $\rightarrow +\infty$ as $x \rightarrow$ the i -th face.

Gradient & Hessian

With $s_i(x) := b_i - a_i^\top x$ (slack):

$$\nabla F(x) = \sum_{i=1}^m \frac{a_i}{s_i(x)}$$

$$\nabla^2 F(x) = \sum_{i=1}^m \frac{a_i a_i^\top}{s_i(x)^2}$$

Analytic Center

The unique x^* with $\nabla F(x^*) = \mathbf{0}$ is called the **analytic center** of P .

Physical view: equilibrium of m repulsive forces.

Applications:

- Assessing volume of P
- Polynomial-time LP** (Renegar 1988)

Complexity

$$\nabla F: O(nm); \quad \nabla^2 F: O(n^2m)$$

Sparse a_i : both reduce to $O(m)$

Graph Laplacian structure: $\tilde{O}(m)$

[Spielman–Teng]

Theorem 2: Quadratic Convergence (Euclidean Norm)

Theorem (Quadratic Convergence w.r.t. Euclidean Norm)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, x^* its minimizer, x_0 any starting point, $x_1 := x_0 + n(x_0)$. If condition **NE**(M) holds, then

$$\|x_1 - x^*\|_2 \leq M \|x_0 - x^*\|_2^2$$

Condition NE (Newton-Euclidean)

$\exists$ ball $B(x^*, R) \ni x_0$ and constants $h, L > 0$ with $M \geq \frac{L}{2h}$ such that:

- ① $\|H(x)^{-1}\| \leq \frac{1}{h}$ for all $x \in B(x^*, R)$ (large $h \Rightarrow$ large Hessian)
- ② $\|H(x) - H(y)\| \leq L \|x - y\|_2$ for all $x, y \in B(x^*, R)$ (Hessian is L -Lipschitz)

Proof Strategy

Using the Fundamental Theorem of Calculus:

$$x_1 - x^* = H(x_0)^{-1} \int_0^1 (H(x_0 + t\delta) - H(x_0)) \delta dt$$

where $\delta := x^* - x_0$. Then:

$$\|x_1 - x^*\|_2 \leq \frac{L \|H(x_0)^{-1}\|}{2} \|x^* - x_0\|_2^2$$

Limitations: Why Euclidean Norm Fails

Failure Example

Polytope $P = [-K_1, K_1] \times [-1/K_2, 1/K_2] \subseteq \mathbb{R}^2$,
 $K_1, K_2 \gg 1$ (very elongated).

- At $x^* = (0, 0)$: $h \leq 2/K_1^2$ (tiny!)
- Lipschitz constant: $L \approx K_2^2$ (huge!)
- $M = L/(2h) = \Omega(K_1^2 K_2^2)$

Theorem 2 gives **no useful bound**, yet
Newton's method converges rapidly in practice!

Affine Invariance of Newton's Method

Under any affine map $\varphi(x) = Ax + b$:

- Newton's *trajectory is identical* in both x - and y -coordinates
- Gradient descent **does not** share this property

Problem with Theorem 2: h and L change under reparametrisation $\Rightarrow$ the bound is *not affinely invariant*.

Solution

Replace the Euclidean norm with the **local norm** induced by the Hessian.

Newton's Method as Gradient Descent on a Riemannian Manifold

Motivating Example

Minimise $f(x_1, x_2) = x_1^2 + Kx_2^2$, $K \gg 1$.

- Gradient descent with fixed η : overshoots ($\eta \approx 1/K$ needed)
- Use instead the *Hessian norm*:
 $\|(u_1, u_2)\|_o = \sqrt{u_1^2 + Ku_2^2}$
- Steepest descent in $\|\cdot\|_o$: $u_{\text{opt}} \propto -x$, points *directly* to $x^* = 0$
- For $\eta = 1$: one step finds the exact minimum!

Local Inner Product & Local Norm

For strictly convex f with $H(x) = \nabla^2 f(x)$:

$$\langle u, v \rangle_x := u^\top H(x) v$$

$$\|u\|_x := \sqrt{u^\top H(x) u}$$

These define a **Riemannian metric** on $\mathbb{R}^n$, varying smoothly with x .

$\Rightarrow$ a **Hessian manifold**.

Newton's Flow

Steepest descent w.r.t. $\|\cdot\|_x$:

$$u_{\text{opt}} \propto -H(x)^{-1} \nabla f(x) = n(x)$$

Theorem 4: Convergence in the Local Norm

Potential Function (affinely invariant)

$$\lambda(x) := \|n(x)\|_x = \sqrt{\nabla f(x)^\top H(x)^{-1} \nabla f(x)}$$

Interpretation: $\frac{1}{2}\lambda(x)^2 = \text{gap between } f(x) \text{ and minimum of its local quadratic approximation.}$

Theorem (Quadratic Convergence w.r.t. Local Norm)

Let f satisfy condition **NL**, $x_0 \in \mathbb{R}^n$, $x_1 := x_0 + n(x_0)$.

$$\lambda(x_0) < \frac{1}{6} \implies \lambda(x_1) \leq 3\lambda(x_0)^2$$

Condition NL (Newton-Local) — affinely invariant

For all x, y with $\|y - x\|_x = \delta < 1$:

$$(1 - 3\delta)H(x) \preceq H(y) \preceq (1 + 3\delta)H(x)$$

Proof Key Steps

- 1 **Lemma 6** (low distortion): $\|y - x\|_x \leq \frac{1}{6} \implies \frac{1}{2}\|\cdot\|_x \leq \|\cdot\|_y \leq 2\|\cdot\|_x$
- 2 Write $\nabla f(x_1) = M(x_0)H(x_0)^{-1}\nabla f(x_0)$
... etc

Comparison of Convergence Analyses

Property	Theorem 2 (Euclidean)	Theorem 4 (Local Norm)
Progress measure	$\ x_1 - x^*\ _2 \leq M \ x_0 - x^*\ _2^2$	$\lambda(x_1) \leq 3\lambda(x_0)^2$
Condition	NE: ball, constants h, L	NL: local Lipschitz on H
Affinely invariant	No — h, L change with coords	Yes — all quantities invariant
Log-barrier failure	Yes — $M = \Omega(K_1^2 K_2^2)$	No — bounded for any P
Proof tool	Lipschitz H + operator norm	FTC + PSD ordering + Lemma 6

The Right Choice of Norm Matters

The Euclidean norm is a poor fit for Newton's method, which is intrinsically coordinate-free. The local (Hessian) norm restores affine invariance and gives a clean, universally applicable bound.

Key Takeaways

- ① **Doubly-exponential convergence.** Near a root, $k \approx \log \log(1/\varepsilon)$ iterations suffice — vastly better than first-order methods.
- ② **Second-order method.** Each step minimises the local quadratic approximation of f . One step is exact for strictly convex quadratics.
- ③ **Cost–benefit trade-off.** Requires solving an $n \times n$ linear system per step ($O(n^3)$ general; $\tilde{O}(m)$ for Laplacians).
- ④ **Affine invariance.** Newton's trajectory is coordinate-free. The local norm captures this correctly; the Euclidean norm does not.
- ⑤ **Connection to interior-point methods.** Logarithmic barrier + Newton's method yields a polynomial-time LP algorithm (Renegar 1988). Condition $NL \approx \textit{self-concordance}$ (Nesterov–Nemirovskii 1994).

References I

 A. Galntai.
The theory of Newton's method.
J. Comput. Appl. Math., 124(1):25–44, 2000.

 Y. Nesterov and A. Nemirovskii.
Interior-Point Polynomial Algorithms in Convex Programming.
SIAM, 1994.

 J. Raphson.
Analysis aequationum universalis.
London, 1690.

 J. Renegar.
A polynomial-time algorithm, based on Newton's method, for linear programming.
Math. Program., 40(1–3):59–93, 1988.



D. A. Spielman and S.-H. Teng.

Nearly-linear time algorithms for graph partitioning, graph sparsification, and solving linear systems.
In *STOC 2004*, pages 81–90, 2004.